Int. J. Production Economics 126 (2010) 204–211

Contents lists available at ScienceDirect

Int. J. Production Economics

journal homepage: www.elsevier.com/locate/ijpe

The sufficiency of target costing for evaluating production-related decisions

Robert Kee *

Culverhouse School of Accountancy, University of Alabama, P.O. Box 870220, Tuscaloosa, AL 35487-0220, USA

ARTICLE INFO

Article history:
Received 2 July 2009
Accepted 13 March 2010
Available online 19 March 2010

Keywords:
Target costing
Product cost
Production-related decisions

ABSTRACT

Target costing is a cost management system designed to develop products with a level of profitability deemed adequate to justify their production. This paper examines whether production-related decisions made with target costing consistently add economic value to the firm. A traditional target costing model is compared to one that incorporates the cost of capital. An analysis of the two models reveals that the traditional target costing model systematically underestimates the marginal cost of invested funds and overestimates the marginal cost of cash-related production resources. A numerical example and graphical analysis demonstrate that a traditional target costing model can lead to accepting products that have a negative net present value, while rejecting products that have a positive net present value. Mathematical analysis of a traditional target costing model indicates that it is a systemic property of the model.

© 2010 Elsevier B.V. All rights reserved.

1. Introduction

Target costing is a cost management system for developing products whose profitability is deemed sufficient to justify their production. The strategy underlying target costing is that 80–85% of a product's life cycle cost is determined during its development stage. Consequently, target costing focuses its efforts on a product's development since it has the greatest potential for managing a product's cost. Target costing begins with research into the attributes and quality customers want in a prospective product and the price they are willing to pay for these features. The next step involves subtracting the profitability required by the firm to manufacture a product from its market price to determine its target, or allowable, cost. All subsequent efforts of target costing are focused on achieving a product's allowable cost. If a product's cost can be achieved, it is produced. Otherwise, the product is rejected as financially infeasible to manufacture.

In a study of the control issues related to target costing, Kee and Matherly (2006) provide an example of a product whose production cost is less than its target cost, but earns a negative net present value (NPV), while a product whose production cost is greater than its target cost earns a positive NPV. The authors' example is problematic since it suggests that target costing may lead to accepting (rejecting) products that destroy (add) economic value to the firm. Kee and Matherly (2006) do not discuss whether their example reflects a general property of target costing or is simply an artifact of the product cases they developed. If the example used by Kee and Matherly (2006) represents an underlying property of target costing, then it can lead to dysfunctional production-related decisions. Consequently, it is critical to examine the properties of target costing to determine whether the example used by Kee and Matherly (2006) represents an anomaly or is systemic. The purpose of this paper is to examine the sufficiency of target costing for making production-related decisions. Target costing is sufficient if it consistently leads to accepting products for production that add economic value to the firm. The paper identifies the property of target costing that leads to the suboptimal product acceptance and rejection decisions illustrated by Kee and Matherly (2006). Equally important, the paper examines the impact of this property upon the development of products, as well as the conditions that affect its impact upon production-related decisions.

The remainder of this article is organized as follows. Section 2 discusses target costing. Section 3 mathematically models a traditional target costing system and one that incorporates the cost of capital and examines their economic properties. In Section 4, a numerical example and graphical analysis is used to illustrate the economic properties of the target costing models. Section 5 examines the economic loss from product acceptance decisions developed with a traditional target costing model. The summary and conclusions of the article are presented in Section 6.

2. Target costing

The first part of Section 2 discusses product development. The second part reviews relevant target costing research while the third part examines the economic domain of target costing.

* Tel.: +1205 348 2909; fax: +1 205 348 8453.
E-mail address: rkee@cba.ua.edu

0925-5273/$ - see front matter © 2010 Elsevier B.V. All rights reserved.
doi:10.1016/j.ijpe.2010.03.008

2.1. Product development

Target costing begins with marketing research to identify customer expectations of a product's functionality, quality, and price. The market-derived price is reduced by the profitability necessary for the firm to manufacture the product. The difference between a product's market price and desired profitability represents the product's allowable, or target, cost. The measurement of the profitability necessary for a firm to manufacture a product is usually based on return on sales or profit margin metric (Cooper and Slagmulder, 1997; Sakurai, 1989). After a product's target cost has been established, a multidisciplinary team is commissioned to design and to develop the product to meet customer expectations within the constraint of its allowable cost.

Target costing uses a variety of techniques and methodologies to manage product design and cost. One of the primary techniques, value engineering, is utilized to identify the primary and secondary functions of a product. The multidisciplinary team then uses this information to identify innovative means of providing a function and/or providing it at a reduced cost. Teardown analysis and reverse engineering is frequently employed to perform comparative analyses of competitors' products' functionality, quality, cost, and design and production innovations. Functional cost analysis is used to take the value customers assign to each function and allocate a product's target cost to its functions and components (Yoshikawa et al., 1994). The design team applies value engineering to reduce the cost of each function and component whose cost exceeds its allowable cost. For example, the cost of a function may be reduced by finding alternative means for its implementation. Conversely, the cost of a component may be reduced either by using less expensive material and/or using fewer parts. The multidisciplinary team begins to work early with suppliers to receive their input into the design process and to communicate the firm's expectations of purchased components' costs and quality. Finally, value engineering is used to examine the production processes used to manufacture a product and its components to enhance functionality and/or reduce cost. Other techniques, such as Quality Function Deployment, Design for Manufacture and Assembly, and Cost Deployment Flowcharts, are often used in the target costing process.

An integral aspect of target costing and value engineering involves the estimation of a product, its functions, and its components cost. The costs of components used in an older version of the product or components common to other products may be estimated from their current costs less planned cost reductions. The costs of unique components may be estimated from detailed analysis of the production processes required for their production. Parametric cost estimations may be used to estimate a component's cost from historical cost data and the component's statistical relationship to one or more cost drivers (Sullivan et al., 2006). Japanese firms apply detailed cost tables that encode their cost structure to estimate a component's cost (Cooper and Slagmulder, 1997). The cost of a product's function is derived from the estimated cost of the components that provide the function. Similarly, the cost of a product may be developed by aggregating the cost of its components and assembly-related costs.

Target costing is an iterative process. The multidisciplinary team continues to redesign a product and apply value engineering until either its estimated cost is less than or equal to its allowable cost or further cost reductions are no longer feasible. In cases where a product's estimated cost continues to exceed its allowable cost, the product is not developed further. The cardinal rule of target costing requires that only those products whose estimated cost is less than or equal to their target cost be produced (Cooper and Slagmulder, 1997). The cardinal rule is necessary to maintain the discipline of target costing during product development.[1] Once a product enters production, a program of continuous improvement, or Kaizen costing in Japanese firms, is used to reduce costs further and improve product quality (Cooper and Slagmulder, 1997).

2.2. Target costing research

Early target costing researchers used survey methodology to examine the attributes of firms adopting target costing. In surveys of Japanese and Dutch firms, Tani et al. (1994) and Dekker and Smidt (2003), respectively, found that the primary reason cited for using target costing was cost reduction. Other reasons frequently cited included improving product quality, achieving timely introduction of new products, and improving communication (Tani et al., 1994; Ellram, 2000). Analysis of target costing firms revealed that they were often in highly competitive markets (Swenson et al., 2003) with high environmental uncertainty (Dekker and Smidt, 2003). However, in a survey of Swedish manufacturing firms, Ax et al. (2008) found that target cost adoption was negatively correlated with perceived environmental uncertainty. Surveys of target costing adoption indicate that it is widely employed in different countries. For example, Sulaiman et al. (2004) reported target cost usage rates of 35% and 41% for Indian and Malaysian firms, respectively. Similarly, Tani et al. (1994) indicated a target cost adoption rate of 33% among Japanese manufacturing firms.

Applying more sophisticated methodologies, several studies have examined the relationship between target costing and structural aspects of the firm, as well as implementation issues. Hibbets et al. (2003) found that firms with a strategy of product differentiation and strong competition were more likely to use target costing. In a laboratory experiment, Everaert and Bruggeman (2002) noted that having a cost target for new product development led to lower cost without impairing design quality or development time. However, under high time pressure, a cost target led designers to work longer without a corresponding cost decrease (Everaert and Bruggeman, 2002). In a similar setting, Monden et al. (1997) found that participation in setting target costs and evaluations based on a designer's accomplishments were factors that improved cost reduction performance. Based on a simulated laboratory experiment, Akter et al. (1999) reported that difficult, or tighter, targets also tended to improve cost reduction performance.

A final set of studies has examined potential dysfunctional aspects of target costing. As noted earlier, Everaert and Bruggeman (2002) suggested that, under high time pressure, target costing led design engineers to work longer without an accompanying decrease in cost. Kato et al. (1995) suggested that target costing can lead firms to excessive market segmentation, longer product development periods, employee burnout, and organizational conflict. They also noted that target costing can intensify problems with suppliers when cost-reduction demands are passed down to them. Based on observations from field studies, Davila and Wouters (2004) warned that target costing is less useful for products where technology, time-to-market, or demanding customers are more important to a product's success than its cost. They also indicated that target costing may be difficult to implement for firms in complex and uncertain environments.

[1] Cooper and Slagmulder (1997) indicate the cardinal rule may be violated for products with high visibility for the firm, products that use the next generation of technology, or products that fill a critical gap in the firm's product line.

2.3. The domain of target costing

Cooper and Slagmulder (1997) note that target costing focuses primarily on direct costs. However, they also indicate that it can be used to reduce indirect costs. In a case study of Nissan Motor Company, Cooper and Slagmulder (1997) reported that a product's final target cost included indirect manufacturing costs. In a survey of Japanese firms, Tani et al. (1994) indicate that overhead and depreciation on new investments were incorporated into target cost by 80.7% and 83.3% of surveyed firms, respectively.[2]

When a product requires a substantial investment, Cooper and Slagmulder (1997) suggest that target costing requires an adjustment for capital intensity. They propose that a product's profit margin should be set large enough to ensure that the firm earns an adequate profit (Cooper and Slagmulder, 1997). Surveys of firms using target cost do not mention whether the cost of capital is included in a product's target cost (Tani et al., 1994; Dekker and Smidt, 2003). Furthermore, examples used in target costing studies do not include the cost of capital in their computation of a product's target cost (Cooper and Slagmulder, 1997). In a field study of a Brazilian firm, Filomena et al. (2009) stated that the time value of money was not included in target costing, but indicated that it could be.[3] In a second study of a Brazilian firm, Ibuski and Kaminski (2007) consider only variable cost in the computation of target cost. However, the cost of capital was later considered through the use of payback analysis that incorporated the cost of capital. As noted by Afonso et al. (2008), there are alternative approaches to target costing, which can be used to manage cost and reduce time-to-market in product development. Similarly, different firms include different costs in a product's target cost. However, from prior studies, it appears that many firms incorporate direct, as well as indirect, overhead and depreciation costs while few firms include the cost of capital in a product's target cost.

The exclusion of the cost of capital from a product's cost fails to fully measure the cost of the resources used in its production. The cost of capital is a cost of the funds used to create a product. The funds invested in the long-terms assets required to manufacture a product could have been invested elsewhere. The return that could have been earned on alternative investments with similar risk represents the opportunity cost from using the funds. Alfred Marshall, an economist in the 1800s, noted that a firm earns a profit only when its revenue exceeds its production cost and the cost of invested funds. More recently, Stewart (1991) asserts that invested funds must earn a return equal to their cost of capital in order to leave the value of the firm unchanged. In effect, the cost of capital reflects a minimum rate of return the firm must earn in order to engage in an economic activity. Consequently, the cost of capital should be included in a product's cost to enable a firm's management to determine whether it adds or destroys economic value.

One method of incorporating the cost of capital into the cost of a product involves using the concept of economic value added (EVA[4]). EVA is defined as a cost object's net operating profit after taxes less a charge for the cost of capital used to earn the operating profit (Stewart, 1991). EVA represents a one-period measure of a cost object's economic performance. However, a cost object may have a life cycle of several years. Hartman (2000) and Shrieves and Wachowicz (2001) provide mathematical proofs that discounting an investment's EVA over its life at the firm's cost of capital is equivalent to its NPV. Consequently, the discounted EVA of a cost object, such as a product, is equal to its NPV.

EVA provides a means of integrating the cost of capital into a product's target cost. Discounting a product's EVA is a life cycle measure of its economic performance congruent with the principles of financial theory. That is, an investment should only be undertaken if its return is greater than or equal to its cost of capital. A target costing model based on discounted EVA will be referred to as a target costing model incorporating the cost of capital throughout the remainder of this paper. Comparing decisions made with a traditional target costing model and a model incorporating the cost of capital is useful for understanding the differences in their financial consequences. Equally important, the margin cost of the traditional target costing model and one incorporating the cost of capital, provides evidence for understanding their economic properties.

3. Target costing models

To develop and compare a traditional target costing model with one incorporating the cost of capital, the following notation is used:

The variable i is a period index, $i=1, 2,\ldots, N$, N is a product's economic life, P_i is the unit price of a product in period i, C_i is unit production cost in period i, O_i is unit cash-related production cost in period i, O_N is cash-related production cost over a product's life, I_i is the book value of long-term assets used in period i, Q_i is the quantity of a product produced and sold in period i, r_i is the cost of capital in period i, t_i is the effective tax rate in period i, α is the firm's required profit margin percentage for a prospective product, $PV_{N,r}$ is the present value of an annuity for N periods and cost of capital of r, NPV is net present value, NI is net income after taxes, and, NI_N is net income after taxes over a product's life.[5]

The variable C_i represents the cost of the material, labor, and overhead resources used to create a product, while the variable O_i reflects these same costs, but excludes non-cash related resources, such as depreciation expense. C_i and O_i include both variable and fixed costs.[6] The variable α is the firm's required profit margin, a cost object's net income divided by its sales (Ingram and Albright, 2007). The firm's required profit margin for a product includes an allowance for non-production-related costs, such as selling and administrative costs.

Assuming that a product's price, unit production cost, annual sales, and tax rate are constant over its life, the traditional target costing model may be stated

$$NI = PQ(1-t) - CQ(1-t) \tag{1}$$

$$\text{Subject to: } NI/(1-t) \geq \alpha PQ \tag{1 – A}$$

Income in Eq. (1) is measured on an after-tax basis to be comparable with subsequent analysis of a target costing model incorporating the cost of capital. The first and second terms on the right-hand side of Eq. (1) measure a product's expected revenue and production expenses after taxes, respectively. The subject to condition expressed by Eq. (1-A) represents the constraint a

[2] Since direct conversion cost was a separate response in the survey question of what cost are included in target cost management, respondents in the Tani et al. (1994) study appear to be referring to direct as well as indirect overhead.

[3] The cost of capital for long-term assets is commonly included in the analysis of economic decisions, such as design analysis, in engineering economic texts, see for example, Park (2007). Similarly, Kirchain and Field (2001) call for the inclusion of capital cost in process-based cost modeling systems to facilitate the understanding of the cost implications of engineering decisions.

[4] EVA is a registered trademark of Stern Stewart & Company.

[5] If the subscript i is omitted from a variable, then it has the same value each period.

[6] As noted in the prior section, some firms implement target costing based on direct, or variable costs. This paper includes direct, as well as indirect, costs in the development of a product's target cost. Consequently, the models and conclusions developed in this paper will have to be adjusted for firm's including only direct costs in their implementation of target costing.

product must meet in order for the firm's management to justify its production.

Target cost represents the maximum value of C, in Eq. (1), that a product may incur while meeting the constraint of Eq. (1-A). Therefore, substituting the right-hand side of Eq. (1) into the left-hand side of Eq. (1-A) and solving for the maximum value of C results in

$$C = P(1-\alpha) \tag{2}$$

In Eq. (2), C measures a product's allowable cost. In effect, the value of C derived from Eq. (2) becomes the objective that a product's design and production team strives to achieve. Once the design and production team has eliminated as much cost as possible from a product, without compromising the functionality and quality that customers demand, the product's estimated cost is compared with C. Consequently, C developed from Eq. (2) becomes the basis for product acceptance and rejection decisions. Throughout the remainder of the paper, it is assumed that a product's design and production team has reduced a product's cost to the lowest extent possible and that a product's target cost is being used to evaluate whether a prospective product should be manufactured or not.

As noted earlier, Hartman (2000) and Shrieves and Wachowicz (2001) demonstrated that a product's discounted EVA is mathematically equivalent to its NPV. Therefore, a product's life cycle economic income, or NPV, may be expressed

$$NPV = \sum_{i=1}^{N} \frac{(P_i - C_i)Q_i(1-t_i)}{(1+r_i)^i} - \sum_{i=1}^{N} \frac{r_i I(N+1-i)}{N(1+r_i)^i} \tag{3}$$

The first term in Eq. (3) measures a product's operating income after taxes. The second term reflects the cost of capital used to earn the operating income. The original investment in long-term assets, or I, is multiplied by $(N+1-i)/N$ to represent the decline in book value as successive periods' depreciation expenses are taken.[7] Assuming as before, that a product's price, unit production cost, annual product sales, effective tax rate, as well as the cost of capital, are constant over a product's life, Eq. (3) simplifies to:

$$NPV = \frac{(P-C)Q(1-t)[1-(1+r)^{-N}]}{r} - I\left[1 - \frac{1-(1+r)^{-N}}{Nr}\right] \tag{4}$$

$$\text{Subject to: } NPV \geq 0 \tag{4 – A}$$

The first term on the right-hand side of Eq. (4) is the present value of a product's operating income after taxes, while the second term is equal to the difference between the values of the funds initially invested, or I, and the present value of the depreciation expense taken over a product's life. Eq. (4-A) represents the constraint a product must meet for it to be produced, i.e., the product must earn a return equal to or greater than its cost of capital. Eqs. (4) and (4-A) represent the discounted economic income counterparts to accounting profitability expressed in Eqs. (1) and (1-A). In effect, Eqs. (4) and (4-A) could be used to implement target costing the same as Eqs. (1) and (1-A). The only difference between the two models is that target costing, based on Eqs. (4) and (4-A), incorporates the cost of capital as an expense and discounts each period's EVA to the beginning of Period 1.

Replacing each term in Eq. (4) that is equivalent to the present value of an annuity for N periods and a cost of capital of r, with $PV_{N,r}$, Eq. (4) may be rewritten[8]

$$NPV = (P-C)Q(1-t)PV_{N,r} - I(1-PV_{N,r}/N) \tag{5}$$

$$\text{Subject to: } NPV \geq 0 \tag{5 – A}$$

Substituting the right-hand side of Eq. (5) into the left-hand side of Eq. (5-A) and solving for the maximum value of C results in

$$C = \frac{PQ(1-t)PV_{N,r} - I(1-PV_{N,r}/N)}{Q(1-t)PV_{N,r}} \tag{6}$$

In Eq. (6), C equals a product's target cost. When a product's price, annual sales, tax rate, and/or cost of capital are not constant over a product's economic life, Eq. (3) should be used to determine a product's target cost.

3.1. Economic properties of target costing models

A comparison of a product's target cost measured with a traditional target costing model, Eq. (2), and one incorporating the cost of capital, Eq. (6), indicates that their measurements of a product's allowable cost differ substantially. To understand the underlying difference and the product acceptance and rejection decisions they may lead to, consider how the two models integrate the costs of cash-related production costs and capital resources into their measurements of a product's economic performance. In Eq. (1), the basis from which C was determined, a product's production cost per unit is equal to its cash-related production cost per unit and depreciation expense per unit, or $C = O + I/QN$. Substituting $(O + I/QN)$ for C in Eq. (1), it can be rewritten

$$NI = PQ(1-t) - OQ(1-t) - I(1-t)/N \tag{7}$$

$$\text{Subject to: } NI/(1-t) \geq \alpha PQ \tag{7 – A}$$

The income over a product's economic life is equal to Eq. (7) multiplied by the number of periods a product is expected to be produced, or N. Replacing a product's total cash-related production cost with O_N, and its lifetime profitability with NI_N, Eqs. (7) and (7-A) may be stated

$$NI_N = NPQ(1-t) - O_N(1-t) - I(1-t) \tag{8}$$

$$\text{Subject to: } NI_N/(1-t) \geq \alpha NPQ \tag{8 – A}$$

The marginal cost of cash-related production costs and capital resources are the first derivative of Eq. (8) with respect to these two variables. The incremental cost of a dollar of cash-related production costs and invested funds is their after-tax cost, or $(1-t)$ for both variables. Therefore, the incremental cost of an additional dollar of cash-related production resources and invested funds each reduces profitability by an equivalent amount. In effect, a dollar decrease in cash-related production costs could be replaced with a dollar increase in invested funds, while leaving the profitability of a product unchanged. Consequently, a product's cash-related production costs and invested funds are perfect cost substitutes for each other.

Eq. (5), like Eq. (1), can also be rewritten to differentiate between cash-related production costs and invested funds. Substituting $(O + I/QN)$ for C in Eq. (5), it may be expressed

$$NPV = PQ(1-t)PV_{N,r} - OQ(1-t)PV_{N,r} - I(1-t)PV_{N,r}/N - I(1-PV_{N,r}/N) \tag{9}$$

$$\text{Subject to: } NPV \geq 0 \tag{9 – A}$$

Replacing annual cash-related production costs, or OQ, with O_N/N, in Eq. (9), and combining its last two terms, the equation may be rewritten

$$NPV = PQ(1-t)PV_{N,r} - (O_N/N)(1-t)PV_{N,r} - I(1-(t\,PV_{N,r}/N)) \tag{10}$$

$$\text{Subject to: } NPV \geq 0 \tag{10 – A}$$

To determine the marginal cost of cash-related production resources and invested funds, the first derivative of Eq. (10) is

[7] The equations and examples used throughout the paper assume that straight-line depreciation is used.

[8] The last term in Eq. (5) is the original investment less the present value of the annual depreciation expense. Consequently, when the last term in Eq. (5) is subtracted from the present value of net income, it is the same as transforming net income from an accrual to a cash flow basis.

taken with respect to these two variables. The resulting incremental cost of cash-related production resources and invested funds is

$$d(NPV)/dO_N = -((1-t)\ PV_{N,r})/N \qquad (11-\mathrm{A})$$

$$d(NPV)/dI = -(1-(t\ PV_{N,r}/N)) \qquad (11-\mathrm{B})$$

An examination of Eq. (11-A) reveals that it represents the after-tax discounted value of a dollar of cash-related production resources incurred uniformly over a product's life. Conversely, Eq. (11-B) equals the cost of a dollar of funds invested at the beginning of the period in which a product is produced, less the present value of the depreciation tax savings generated each period over the product's life.

The relationship between the marginal cost of a dollar of cash-related production resources and a dollar of invested funds under a traditional target cost and a model incorporating the cost of capital is[9]

$$\Big((1-t)\ PV_{N,r}\Big)/N \le (1-t) \le \Big(1-(t\ PV_{N,r}/N)\Big) \qquad (12)$$

Where

$((1-t)\ PV_{N,r})/N$ is the marginal cost of a dollar of cash-related production expenditures incurred uniformly over a product's life in a target costing model incorporating the cost of capital, $(1-t)$ is the marginal cost of a dollar of cash-related production expenditures and the cost of a dollar of invested funds in a traditional target costing model, and $(1-(tPV_{N,r}/N))$ is the marginal cost of a dollar of invested funds in a target costing model incorporating the cost of capital.

The mathematical relationships expressed in Eq. (12) arise from the numeric domain of $PV_{N,r}/N$. For values of r greater than zero, $PV_{N,r}/N$ will always be less than one. The first term in Eq. (12) multiplies $(1-t)$ by $PV_{N,r}/N$. Therefore, the second term in Eq. (12), or $(1-\mathrm{t})$, is always greater than or equal to the first term, which consists of $(1-\mathrm{t})$ multiplied by $PV_{N,r}/N$. The third term in Eq. (12) multiples the tax rate, t, times $PV_{N,r}/N$. Therefore, the variable t in the second term is greater than or equal to $tPV_{N,r}/N$ in the third term. Consequently, when t and $tPV_{N,r}/\mathrm{N}$ are subtracted from 1, the third term will always be greater than or equal to the second term in Eq. (12).

The significance of Eq. (12) is that the marginal cost of a dollar of cash-related production resources in a traditional targeting costing model is greater than or equal to the marginal cost of a dollar of cash-related production resources incurred uniformly in a target costing model incorporating the cost of capital. Conversely, the marginal cost of a dollar of invested funds in a traditional targeting costing model is less than or equal to the marginal cost of a dollar of invested funds in a target costing model incorporating the cost of capital. The traditional target costing model overestimates the marginal cost of cash-related production resources due to its failure to recognize the time value of money. Conversely, the traditional target costing model underestimates the marginal cost of invested funds due to its failure to include the opportunity cost of capital resources.

Underestimating the incremental cost of invested funds in a traditional target costing model encourages managers to overutilize capital, relative to cash-related production resources, in the design and production of prospective products. Managers have an incentive to use invested funds as long as their marginal benefit is equal to or greater than their marginal cost. Since a traditional target costing model understates the marginal cost of invested funds, more capital will be used than the economic optimal amount. Second, understating the marginal cost of invested funds used in manufacturing a product leads to underestimating its cost. Consequently, a product's profit margin is overstated and thus is more likely to be accepted for production when evaluated with a traditional target costing model. However, when the opportunity cost of invested funds is incorporated into a product's cost, the rate of return earned by the product may be less than the firm's cost of capital. Consequently, the traditional target costing model can lead managers to accept products that destroy, rather than create, economic value for the firm.

Conversely, overestimating the marginal cost of cash-related production resources encourages managers to underutilize this type of resource. Managers will use cash-related production resources as long as their marginal benefit is greater than or equal to their marginal cost. Since the marginal cost of cash-related production resources is overstated in a traditional target costing model, less of this resource will be used in a product's production than the optimal economic amount. Overestimating the marginal cost of cash-related production resources can lead to overestimating a product's cost. Therefore, managers may reject products whose apparent profit margin is less than that required by the firm. However, when the incremental cost of cash-related production resources is correctly measured, the rate of return on these products may be greater than the firm's cost of capital. Consequently, the traditional target costing model may lead managers to reject products that would create economic value for the firm.

4. Numerical and graphical analysis

To illustrate the prior assertions, consider how Products X and Y in Table 1 would be evaluated with a traditional target costing model and one incorporating the cost of capital. Panel I lists the price, annual sales, cash-related production costs, investment, economic life, required profit margin, cost of capital, and tax rate of each product. Notice that Product X and Product Y differ only in their cash-related production resources and invested funds. In Panel II, the estimated unit cost of each product is computed. Assuming straight-line depreciation stated on a per unit basis, the estimated costs for Product X and Product Y total \$107 and \$109, respectively. Allowable cost for a traditional target costing model is computed for Products X and Y using Eq. (2), which equals \$108 per unit for both products. As indicated in Panel II, the estimated unit cost of Product X is less than its allowable, or target, cost, while Product Y's estimated cost is greater than its target cost. Accordingly, a traditional target costing model would accept Product X and reject Product Y for production.

In Panel III, an allowable cost for Products X and Y is computed for a target costing model incorporating the cost of capital. The parameters listed in Panel I of Table 1 were entered into Eq. (6) and solved for the value of C. The resulting target costs for Product X and Product Y were \$93.28 and \$110.43 per unit, respectively. As indicated in Panel III, the estimated unit cost for Product X is greater than its target cost, while the estimated unit cost of Product Y is less than its target cost. Consequently, a target costing model incorporating the cost of capital leads to rejecting Product X and accepting Product Y for production.

To assess the financial consequences of the conflicting product acceptance and rejection decisions listed in Panels II and III of Table 1, the NPV of each product is computed using discounted

[9] Under the traditional target costing model, a dollar of cash production resources incurred today is equivalent to a dollar of these resources incurred at any point in the future or with a dollar of cash production resources incurred uniformly over a product's life. Consequently, comparing the cost of a dollar of cash production resources incurred uniformly over a product's life for a target costing model incorporating the cost of capital is equivalent to comparing any pattern of a dollar of cash production cost under a traditional target costing model. A dollar of invested funds is assumed to occur at the beginning of a product's life under both target costing models.

Table 1
Product decisions using a traditional and a target costing model incorporating the cost of capital.

		X		Y
Panel I: Product Attributes				
Price per Unit (P)	$	120	$	120
Annual Sales in Units (Q)		100,000		100,000
Cash Production Cost per Unit (O)	$	40	$	85
Investment (I)	$	33,500,000	$	12,000,000
Economic Life in Years (N)	$	5	$	5
Required Profit Margin (α)		10%		10%
Cost of Capital (r)		10%		10%
Tax Rate (t)		20%		20%
Panel II: Traditional Target Costing				
Cash Production Cost per Unit (O)	$	40.00	$	85.00
Depreciation Cost per Unit (I/NQ)	$	67.00	$	24.00
Estimated Unit Cost (C)	$	107.00	$	109.00
Target Cost-Eq. (2)	$	108.00	$	108.00
Product Production Decision		Accept		Reject
Panel III: Target Costing Incorporating Cost of Capital				
Cash Production Cost per Unit (O)	$	40.00	$	85.00
Depreciation Cost per Unit (I/NQ)	$	67.00	$	24.00
Estimated Unit Cost (C)	$	107.00	$	109.00
Target Cost-Eq. (6)	$	93.28	$	110.43
Product Production Decision		Reject		Accept
Panel IV: Net Present Value				
Product Revenue	$	12,000,000	$	12,000,000
Cash Operating Expenses	$	(4,000,000)	$	(8,500,000)
Depreciation Expense	$	(6,700,000)	$	(2,400,000)
Operating Income	$	1,300,000	$	1,100,000
Tax Expense (20%)	$	(260,000)	$	(220,000)
Net Income	$	1,040,000	$	880,000
Annual Cash Flow Years 1–5[a]	$	7,740,000	$	3,280,000
Initial Investment	$	(33,500,000)	$	(12,000,000)
Net Present Value of Cash Flows	$	(4,159,310)	$	433,781

[a] Annual net income plus annual depreciation expense.

cash flows in Panel IV. The annual net income for each product is derived from the data in Panel I. Each year's cash flow for Products X and Y is determined by adding depreciation expense back to each product's net income. The NPV of the initial investment and annual cash flows for 5 years, at a cost of capital of 10% for each product, is given at the bottom of Panel IV. As indicated, Product X is expected to provide an NPV of − $4,159,310, while Product Y is expected to generate an NPV of $433,781. As indicated in Panel II, the traditional target costing model leads to accepting Product X, even though it destroys economic value of $4,159,310, and leads to rejecting Product Y, even though it adds an economic value of $433,781.[10] In effect, the traditional target costing model leads to suboptimal product decisions, similar to the example illustrated in the Kee and Matherly (2006) study.

To gain additional insights into the conflicting decisions in Panels II and III of Table 1, Fig. 1 illustrates a traditional and a target costing model incorporating the cost of capital. The horizontal and vertical axes of the graph represent the quantities of cash-related production costs per unit and invested funds used to produce Product X, respectively. The line denoted with the diamond symbol identifies the combination of invested funds and cash-related production costs per unit for Product X that equals its allowable cost under a traditional target costing model. The observations for the traditional target cost line are computed using the data listed for Product X in Panel I of Table 1 and the relationships expressed in Eqs. (8) and (8-A). Product X's lifetime allowable cost is computed using its revenue of $60,000,000 less a required profit margin of 10%. Therefore, any mix of invested funds and cash-related production costs that equals $54,000,000 will yield an allowable cost of $108 per unit for Product X. Cash-related production costs per unit, ranging from $0 to $120 per unit, in increments of $5, are entered into this relationship and solved for the level of invested funds.

The line denoted with the square symbols in Fig. 1 represents the quantities of invested funds and cash-related production costs per unit for Product X that equal its allowable cost under a target costing model incorporating the cost of capital. Each pair of observations comprising the line is computed using Eqs. (10) and (10-A). Cash-related production costs per unit, ranging from $0 to $120, at intervals of $5, are entered into Eq. (10) and solved for the quantity of invested funds that yielded an NPV of zero.[11]

The lines illustrated in Fig. 1 are isoprofit lines. That is, the line for the traditional and the target costing model incorporating the cost of capital represents the mix of invested funds and cash-related production cost per unit that are expected to provide a profit margin of 10% and rate of return equal to 10%, respectively. Consequently, the line for each target costing model and all points below it represents the combinations of invested funds and cash-related production costs per unit for Product X that generate an estimated unit cost either less than or equal to its target cost. Therefore, the area bounded by each line and the vertical and horizontal axes in Fig. 1 represents Product X's acceptance region. Conversely, the area above each line in Fig. 1 defines Product X's rejection region.

An examination of Fig. 1 indicates that the two lines intersect at about invested funds and cash-related production cost per unit of $15,000,000 and $78, respectively. At their point of intersection, the product acceptance and product rejection regions for a traditional and a target costing model incorporating the cost of capital are identical. To the left of the point of intersection, the traditional target costing line lies above the line for the target costing model incorporating the cost of capital. Therefore, the area between the two lines, to the left of their point of intersection, identifies the mix of invested funds and cash-related production costs per unit for Product X that would be accepted by a traditional target costing model, but would be rejected by a target costing model incorporating the cost of capital. In effect, the area between the two lines identifies the mixes of cash-related production costs per unit and invested funds that would be accepted by a traditional target cost model, but earn less than the firm's cost of capital.

To the right of their intersection point, the line for the target costing model incorporating the cost of capital lies above the traditional target costing line. The region between the two lines, to the right of their point of intersection, distinguishes the pairs of invested funds and cash-related production costs per unit for Product X that are accepted by a target costing model incorporating the cost of capital, but rejected by a traditional target costing model. In effect, the area between the two lines in Fig. 1 reflects the quantities of invested funds and cash-related production costs per

[10] Since the revenue of Product X and Product Y is identical, the divergence in their net incomes and NPV illustrated in Panel IV of Table 1 is a result of their different cost structures and how net income and NPV are measured. Product X has lower cash-related production costs and higher investment costs relative to Product Y. Accounting treats these two types of costs as equivalent for computing net income. Since the total of cash-related production and investment costs for Product X are less than those of Product Y, Product X will have a higher net income. Conversely, the NPV measured, based upon discounted EVA, includes the cost of capital as an expense. Since the invested funds of Product X exceed those of Product Y, it will have higher capital costs. When the cost of cash-related production resources, invested funds, and capital cost are discounted, the costs of Product X exceed those of Product Y. Consequently, Product X will have a lower NPV relative to Product Y.

[11] O_N in Eq. (10) was replaced with OQN.

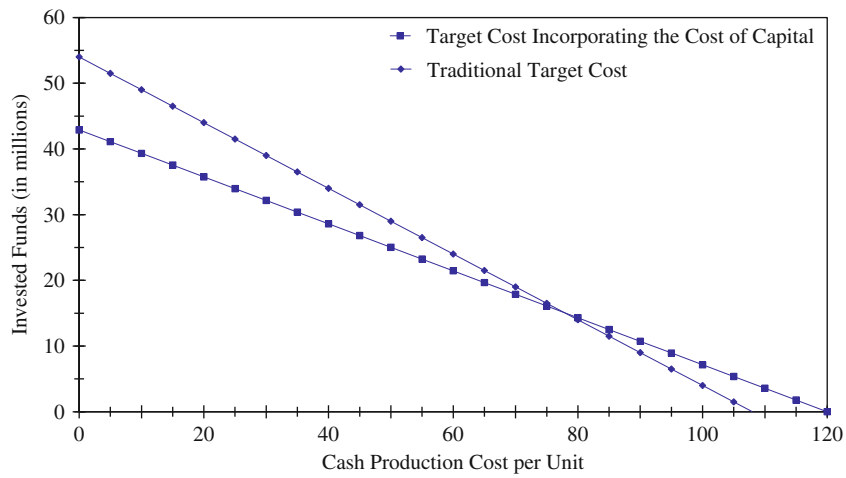

Fig. 1. Product X's acceptance and rejection regions.

unit for Product X that the traditional target costing model would lead to suboptimal product acceptance and rejection decisions. Consequently, the conflicting acceptance and rejection decisions illustrated in Panels II and III of Table 1 represent examples of a much broader domain in which the traditional target costing model is inadequate for evaluating prospective products.

As noted earlier, the lines illustrated in Fig. 1 are isoprofit lines. Therefore, the slope of each line in Fig. 1 is the rate at which the cost of one input can be substituted for the cost of another input and still leave a product's profitability unchanged. The marginal rate of cost substitution of invested funds for cash-related production costs for a traditional target costing model equals 1. Consequently, the slope of the traditional target cost line for total production cost is -1.

For an isoprofit line of a target costing model incorporating the cost of capital, the *NPV* term in Eq. (10) is a constant. Eq. (10) may therefore be restated with I as the dependent variable and O_N as the independent variable. The slope of the resulting equation is

$$dI/dO_N = -\frac{PV_{N,r} - tPV_{N,r}}{N - tPV_{N,r}} \tag{13}$$

In Eq. (13), for $r > 0$, $N > PV_{N,r} > tPV_{N,r}$. Therefore, when $tPV_{N,r}$, is subtracted from $PV_{N,r}$ in the numerator and from N in the denominator, dI/dO_N is negative and less than 1. Accordingly, an isoprofit line for a traditional and a target costing model incorporating the cost of capital cannot have the same slope, regardless of the values for a product's profit margin and cost of capital. Hence, the isoprofit lines for a traditional and a target costing model incorporating the cost of capital cannot be parallel to each other. If $r > 0$, isoprofit lines for a traditional and for a target costing model incorporating the cost of capital will always have conflicting product acceptance and rejection regions. Cooper and Slagmulder (1997) suggest that the profit margin or α used in setting a product's target cost should be increased for capital-intense products. However, as noted earlier, a traditional and a target cost model incorporating the cost of capital cannot have the same slope. Consequently, increasing a product's required profit margin to reflect the cost of capital for products that are capital-intense will still lead to suboptimal product acceptance and rejection decisions, as illustrated in Fig. 1.

5. Economic loss of a traditional target costing model

The vertical distance between the isoprofit lines in Fig. 1 can be used to determine the economic loss from using a traditional target costing model. For a given level of cash-related production cost per unit (the horizontal axis in Fig. 1), the traditional target costing model and one incorporating the cost of capital have different levels of invested funds. The isoprofit line for the target costing model incorporating the cost of capital in Fig. 1 represents the combination of cash-related production costs and invested funds that generate an NPV of zero. Therefore, the NPV of observations on the traditional target costing line is a function of the difference in the invested funds between a traditional and a target costing model incorporating the cost of capital for the same level of cash-related production cost. Eq. (11-B) measures the incremental impact of a dollar of invested funds upon a product's NPV for a target costing model incorporating the cost of capital. Product X has a five year economic life, a cost of capital of 10%, and a tax rate of 20%. Using Eq. (11-B), an incremental dollar of invested funds changes a product's NPV by 0.8484, or 85%. Therefore, the NPV of observations on the traditional target costing line in Fig. 1 equals the difference in its level of invested funds, relative to the model incorporating the cost of capital, times 0.85. Consequently, the greater the distance between the invested funds of a traditional and a target costing model incorporating the cost of capital for a given level of cash-related production costs, the greater the economic loss resulting from using a traditional target costing model.

The profit margin used in a traditional target costing model is endogenously determined. Conversely, the cost of capital employed in the target costing model incorporating the cost of capital is a function of exogenous factors.[12] In the short run, managers have discretionary power over a product's required profit margin, but not over its cost of capital. Consequently, much of the economic loss resulting from using a traditional target costing model is derived from the specific level of the profit margin selected for evaluating a product.

However, the length of a product's economic life along with its cost of capital impact the magnitude of an economic loss incurred from using a traditional target costing model. As the values of N and r for a product decrease, the value of $PV_{N,r}$ begins to approach N as an upper limit. Consequently, the slope of an isoprofit line for a target costing model incorporating the cost of capital, measured with Eq. (13), approaches that of a traditional target costing model. Similarly, the *NPV* measured with Eq. (10) begins to approximate the net income measured with Eq. (8).[13] Therefore,

[12] In the capital asset pricing model, a product's cost of capital is determined by the risk-free rate of return and the product's sensitivity to market risk times the market risk premium.

[13] For a cost of capital equal to zero, the NPV expressed in Eq. (10) decomposes into net income measured with Eq. (8).

as the values for N and r decrease, the isoprofit lines for a traditional and a target costing model incorporating the cost of capital become increasingly closer and increasingly more parallel to each other. As the values of N and r decrease, the difference in the quantity of invested funds between a traditional and a target costing model incorporating the cost of capital becomes smaller. Consequently, as the values for N and r diminish, the economic loss from using a traditional target costing model decreases for any specific profit margin selected by a firm's management.

Alternatively, as a product's economic life and cost of capital increase, $PV_{N,r}$ becomes progressively smaller, relative to N. Similarly, the NPV developed from Eq. (10) declines rapidly, relative to the value of NI_N computed with Eq. (8). As the values of N and r increase, isoprofit lines for a traditional and a target costing model incorporating the cost of capital become increasingly further apart, while the differences in their slopes increase. Therefore, as the values of N and r increase, the difference in the quantity of invested funds between the two target costing models increases. Accordingly, the economic loss incurred from using a traditional target costing model increases for any given profit margin selected by a firm's management.

6. Summary and conclusions

Target costing is a cost management system that incorporates market demands and constraints into a product's development. The firm begins with the price that customers are willing to pay for a product and deducts the required profit margin to determine the product's target, or allowable, cost. All subsequent efforts of target costing are focused on achieving this cost. The cardinal rule of target costing is that only those products that can be produced at a cost less than or equal to their allowable cost are manufactured (Cooper and Slagmulder, 1997).

One of the deficiencies of target costing involves its failure to incorporate the cost of capital into production-related decisions. A discounted EVA model was developed to represent a target costing model incorporating the cost of capital. Analysis of the marginal cost of cash-related production and capital resources with a traditional and a target costing model incorporating the cost of capital revealed that the traditional target costing model systematically underestimates the cost of invested funds, while overestimating the cost of cash-related production resources. A numerical example and graphical analysis demonstrate how a traditional target costing model leads to accepting products with a negative NPV, while rejecting products with a positive NPV. Mathematical analysis reveals that this is a systematic problem rather than an anomaly. The economic loss from production-related decisions made with a traditional target costing model is a function of the profit margin chosen by a firm's management as well as a product's economic life and cost of capital. Therefore, these findings indicate that a traditional target costing model is not always sufficient for production-related decisions.

The conclusions of this paper suggest several avenues for future target costing research. First, analysis of whether firms adjust production-related decisions for products with a long economic life and high cost of capital provides evidence to assess whether firms avoid situations where target costing is most susceptible to suboptimal decisions. Second, an investigation into whether firms modify decisions developed from target costing with information from related capital budgeting decisions would provide insights into whether managers avoid dysfunctional aspects of target costing. Finally, an evaluation of the weight managers assign to a product's target cost in production-related decisions would be useful for determining whether other factors mitigate deficiencies of target cost.

References

Akter, M., Lee, J., Monden, Y., 1999. Motivational impacts of the type and tightness of target cost information: A laboratory experiment. Advances in Management Accounting 8, 159–171.

Afonso, P., Nunes, M., Paisana, A., Braga, A., 2008. The influence of time-to-market and target costing in the new product development success. International Journal of Production Economics 115, 559–568.

Ax, C., Greve, J., Nilsson, U., 2008. The impact of competition and uncertainty on the adoption of target costing. International Journal of Production Economics 115, 92–103.

Cooper, R., Slagmulder, R., 1997. Target Costing and Value Engineering. Productivity Press, Portland, OR.

Davila, A., Wouters, M., 2004. Designing cost-competitive technology products through cost management. Accounting Horizons 18 (1), 13–26.

Dekker, H., Smidt, P., 2003. A survey of the adoption and use of target costing in Dutch firms. International Journal of Production Economics 84, 293–305.

Ellram, L., 2000. Purchasing and supply management's participation in the target costing process. Journal of Supply Chain Management 36 (2), 39–51.

Everaert, P., Bruggeman, W., 2002. Cost targets and time pressure during new product development. International Journal of Operations and Production Management 22 (12), 1339–1353.

Filomena, T., Neto, F., Duffey, M., 2009. Target costing operationalization during product development: model and application. International Journal of Production Economics 118, 398–409.

Hartman, J., 2000. On the equivalence of net present value and market value added as measures of a project's economic worth. Engineering Economist 45 (2), 158–165.

Hibbets, A., Albright, T., Funk, W., 2003. The competitive environment and strategy of target costing implementers: evidence from the field. Journal of Managerial Issues 15 (1), 65–81.

Ibuski, U., Kaminski, P.C., 2007. Product development process with focus on value engineering and target-costing: a case study in an automotive company. International Journal of Production Economics 105, 459–474.

Ingram, R., Albright, T., 2007. Financial Accounting: Information for Decisions, sixth ed. Thomson, Mason, Ohio.

Kato, Y., Boer, G., Chow, C., 1995. Target costing: an integrative management process. Journal of Cost Management Spring, 39–51.

Kee, R., Matherly, M., 2006. Decision control of products developed using target costing. Advances in Management Accounting 15, 267–292.

Kirchain, R., Field, F., 2001. Process-based cost modeling: understanding the economics of technical decisions. Encyclopedia of Materials Science & Engineering. Elsevier Science Publications 1718–1727 pp.

Monden, Y., Akter, M., Kubo, N., 1997. Target costing performance based on alternative participation and evaluation methods: a laboratory experiment. Managerial and Decision Economics 18 (2), 113–129.

Park, C., 2007. Contemporary Engineering Economics, Fourth ed. Prentice Hall, Upper Saddle River, NJ.

Sakurai, M., 1989. Target costing and how to use it. Journal of Cost Management 3 (2), 39–50.

Shrieves, R., Wachowicz, J., 2001. Free cash flow (FCF), economic value added (EVA), and net present value (NPV): a reconciliation of variations of discounted-cash flow (DCF) valuation. Engineering Economist 46 (1), 33–51.

Stewart, G., 1991. The Quest for Value: A Guide for Senior Managers. Harper Business, New York, NY.

Sulaiman, M., Ahmad, N., Alwi, N., 2004. Management accounting practices in selected Asian countries: a review of the literature. Managerial Auditing Journal 19 (4), 493–508.

Sullivan, W., Wicks, E., Luxhoj, J., 2006. Engineering Economy, 13th ed. Prentice Hall, Upper Saddle River, NJ.

Swenson, D., Ansari, S., Bell, J., Kim, I., 2003. Best practices in target costing. Management Accounting Quarterly 4 (2), 12–17.

Tani, T., Okano, H., Shimizu, N., Iwabuchi, Y., Fukuda, J., Cooray, S., 1994. Target cost management in Japanese companies: current state of the art. Management Accounting Research 5, 67–81.

Yoshikawa, T., Innes, J., Mitchell, F., 1994. Applying functional cost analysis in a manufacturing environment. International Journal of Production Economics 36, 53–64.